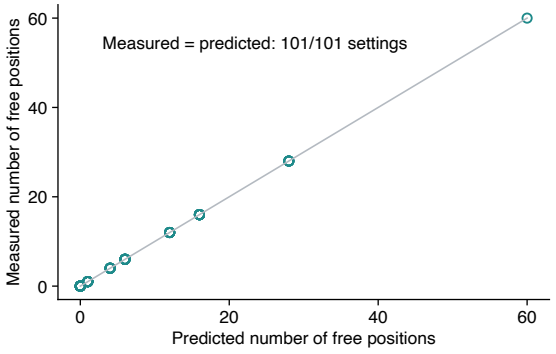
a **Registered: 101 settings (controls included)**



Free-position count: 0, 1, 4, 6, 12, 16, 28, 60
Settings, respectively: 32,14,10,14,10,14,6,1
Free byte positions: arbitrary alternatives jointly inherited with causal roots
Displacement experiment C1; fixed core; 22 parents /101 settings
Includes 22 original controls; markers coincide; diagonal = equality

b **Direct challenges: all passage directions**

Edit / challenge	Intervention cases	Passed
forward / erased	346,290	346,290
forward / joint	6,464	6,464
forward / single	181,050	181,050
reciprocal / erased	486,540	486,540
reciprocal / joint	6,464	6,464
reciprocal / single	40,800	40,800
Total	1,067,608	1,067,608

Z80 C1: 101 settings, 22 parents; actual transfer-8 descendants
Reciprocal = restore displacement; single = 1 site
Joint = payload jointly varied; erased = lost-coordinate control
Pass: exact offspring, ancestry and fixed core
Intervention counts; each tested against 17 partners

c **Z80 C1, L32: lineage 0, transfer 8**

Parent byte 0 = 0x04; copy offset 4 bytes

```
00–15: 04 5e ed b0 04 5e ed b0 04 5e ed b0 04 5e ed b0
16–31: 04 5e ed b0 04 5e ed b0 04 5e ed b0 04 5e ed b0
```

Parent byte 0 = 0x60; copy offset 32 bytes

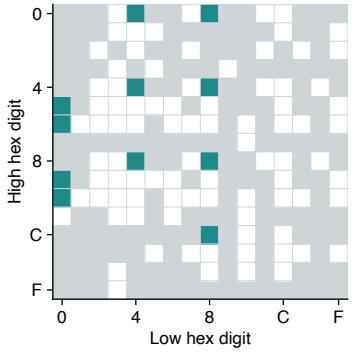
```
00–15: 60 5e ed b0 04 5e ed b0 04 5e ed b0 04 5e ed b0
16–31: 04 5e ed b0 04 5e ed b0 04 5e ed b0 04 5e ed b0
```

Fixed core Ancestry lost Ancestry retained

Byte positions 0–31; each example is one wrapped descendant
Ancestry key refers to original-parent byte sources

d **Exploratory: all 256 first-byte values**

Gain = mean (after – before)/(1 – before); 16 uniform partners
Match to original parent: partner before / offspring after; gen. 2



157: culture-negative (gain <0.3)
88: culture-positive (gain ≥0.3); prefix/offset/phase screen not met
11: culture-positive (gain ≥0.3); prefix/offset/phase screen met

e **Exploratory: sampled-score exceptions**

Measurement	Cells
Motif-bearing cells	2347
Cells with dividing offset	2343
Cells with extra fetched-site scores	53
Cells failing whole-tape match gate	1

53 and 1 are disjoint cell subsets of the 2,343 cells
Extra = fetched-site retention beyond unexecuted sites
Gate deficit: mutant alleles retained at 4 byte positions
Offspring <75% immediate-parent byte match at best cyclic shift

Archived Z80 cohorts G/I/K/L; 5e ed b0 motif (family as c)
2,347 cells from 42 worlds; 4 nondividing excluded

f **Frozen candidate eligibility**

Eligibility / reason	Candidates
nondividing or out of range	4
Eligible	2343

Same motif-cell candidates as e: d divides L, 4≤d≤L
d = copy offset; L = tape length in bytes
Displacement experiment C1: 0/25,856 transfer-record mismatches

101 registered Z80 settings ×16 lineages ×8 transfers ×2 directions
Exact bytes, fixed core and ancestry: each 0/25,856 mismatches